

gynostegium, pollen grains of each theca of an anther are fused into a waxy mass called **pollinium**. The right pollinium of each anther attached to the left pollinium of the adjacent anther by a hair like translator, translator arms (**retinacula**) attached together with the gland like structure called **corpusculum**. Anthers are dithecous, basifixed, introse; dehisce longitudinally.

Gynoecium: Bicarpellary, carpels apically united, superior, 1 to 2 locule with 2 to many ovules in each locule on marginal placentation. Style one and simple, stigma is characteristically thickened, massive and bilobed. A nectariferous disc is often present around or at the base of the gynoecium, (*Catharanthus*).

Fruit: The fruit is variable and can be a berry (*Landolphia*), drupe (*Cerbera*) follicle (*Asclepias*), capsule (*Allamanda*).

Seed: Seeds are endospermous often with crown of hairs.

Botanical description of *Catharanthus roseus*

Habit: Erect ever blooming ornamental plant with milky latex.

Root: Branched tap root system

Stem: Aerial, erect, cylindrical reddish green, glabrous and branched.

Leaves: Usually simple, opposite decussate, exstipulate, subsessile, or petiolate, elliptic – ovate,entire, mucronate, unicostate reticulate venation.

Inflorescence: cymose, axillary pairs.

Flower: Ebracteate,Ebracteolate,subsessile, complete, bisexual, heterochlamydeous, actinomorphic, hypogynous, pentamerous, rosy purple, white or pink.

Calyx: Sepals 5, slightly synsepalous, green showing valvate aestivation.

Corolla: Petals 5, sympetalous, throat of corolla tube hairy forming a corona, twisted (hypocrateriform).

Androecium: Stamens 5, apostemanous, epipetalous, inserted at the mouth of the corolla tube, filaments short, anthers sagittate, dithecous, dorsifixed, introse.

Gynoecium: Bicarpellary, apocarpous, ovaries superior, unilocular, ovules many, placentation marginal, style simple, stigma hour-glass shaped. Two scaly nectaries are present, one on the anterior and another on the posterior side of the ovary.

Fruit: A pair of elongated follicles.

Floral Formula:

$$\text{Ebr.,Ebrl.,} \oplus, \male\female, K_{(5)}, \overset{\frown}{C_{(5)}}, A_5, \underline{G}_{(2)}$$

Economic importance of the family Apocynaceae

S.No	Economic importance	Binomial	Useful part	Uses
1	**Food plant**	*Carissa carandas* (பெரும்களா) *Carissa spinarum* (சிறுகளா)	Fruits	Edible and used in pickles
2	**Medicinal plant**	*Rauvolfia serpentina* (Indian snake root /sarpagandha)	Shoot Dried roots	To treat hypertension and mental disorders Alkaloid (reserpine) obtained from the dried roots, of the plant can lower the blood pressure and used as sedative for patients suffering from *Schizophrenia.*

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